

GateFlow

A Unified Parent-School Platform for Students Arrival and Dismissal

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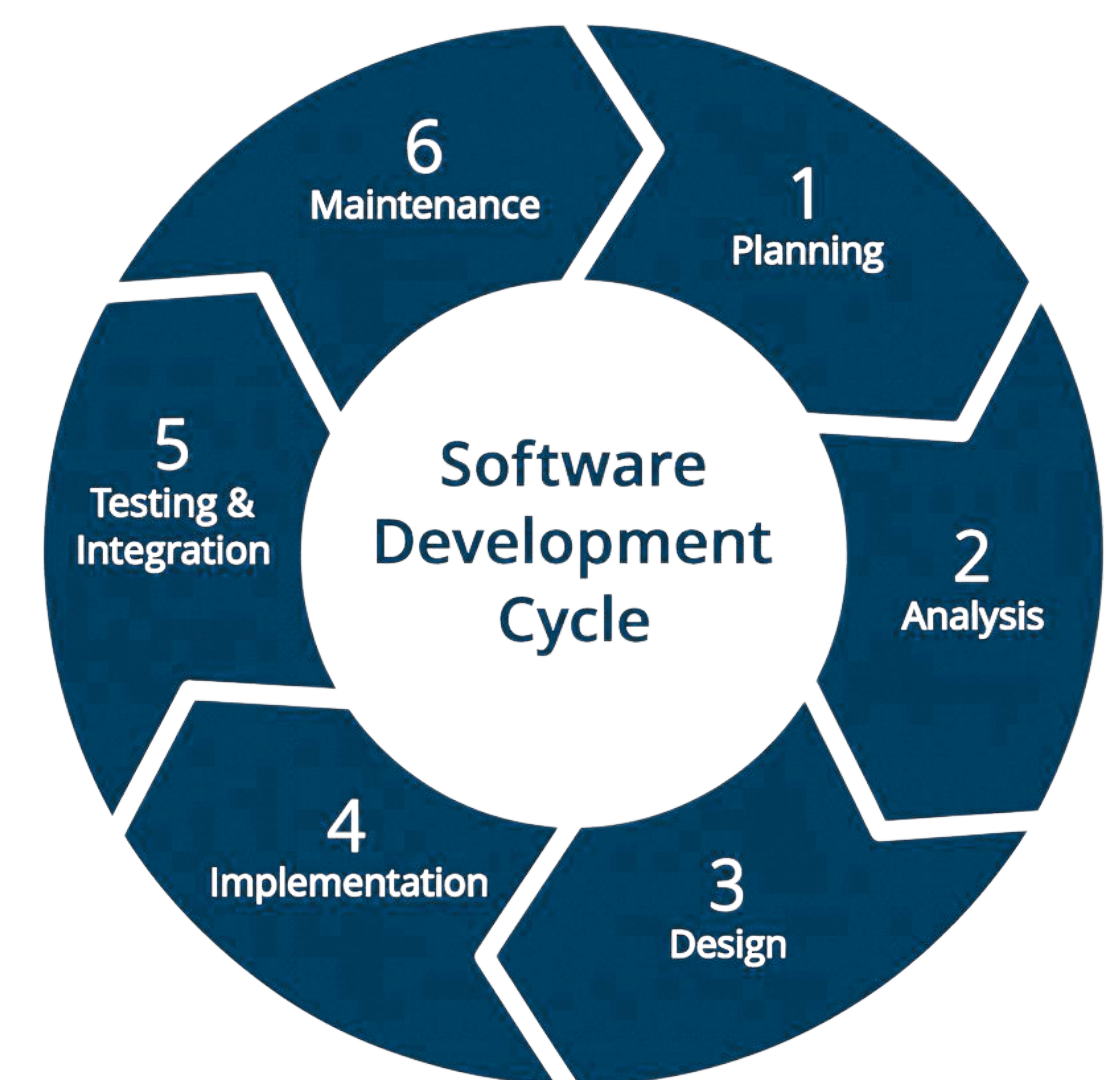
INTRODUCTION

GateFlow is a unified parent-school platform that improves the safety and efficiency of student pickup and drop-off. It addresses common issues such as traffic congestion, long waiting times, and manual coordination by using real-time communication, secure verification, and an organized scheduling process. The system connects parents, school staff, and bus drivers in one organized digital solution.

OBJECTIVES

- Analyze challenges in pickup and drop-off processes (congestion, delays, safety).
- Develop a smart and user-friendly system for managing student dismissal.
- Enhance communication through real-time updates and notifications.
- Improve safety using secure guardian verification.
- Reduce congestion through organized scheduling.
- Support digital transformation by replacing manual processes.

METHODOLOGIES

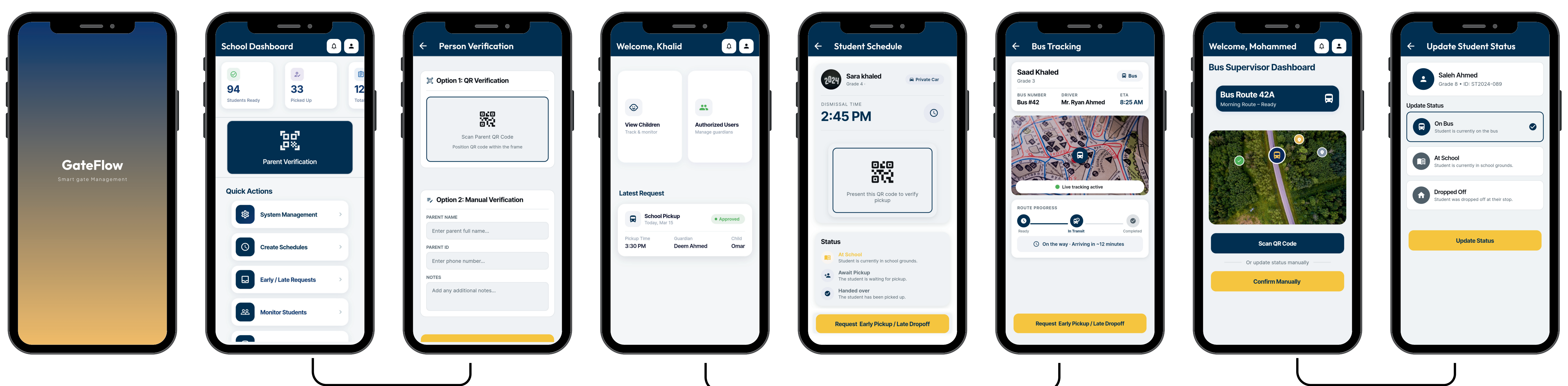


- Planning:**
 - Defined System requirements
- Design:**
 - Activity Diagrams (workflow)
 - ERD (database structure)
- Implementation:**
 - Developed system interfaces
 - Integrated QR code for secure student verification
- Testing:**
 - Performed unit and system testing
 - Verified real-time updates and system functionality

PROBLEM AND MOTIVATION

The manual management of student arrival and dismissal in schools creates significant challenges affecting safety and efficiency. School gates often experience heavy congestion and long waiting times during peak hours, resulting in disorganized queues and increased safety risks. Current authorization methods, such as verbal confirmation or paper forms, are unreliable and prone to error. In addition, communication gaps between school staff, parents, and bus drivers lead to delays and confusion, especially during schedule changes or emergencies. GateFlow is motivated by the need to replace these manual processes with a secure and automated system that organizes vehicle flow and provides real-time updates, ultimately reducing congestion and improving overall safety and coordination within the school environment.

INTERFACES

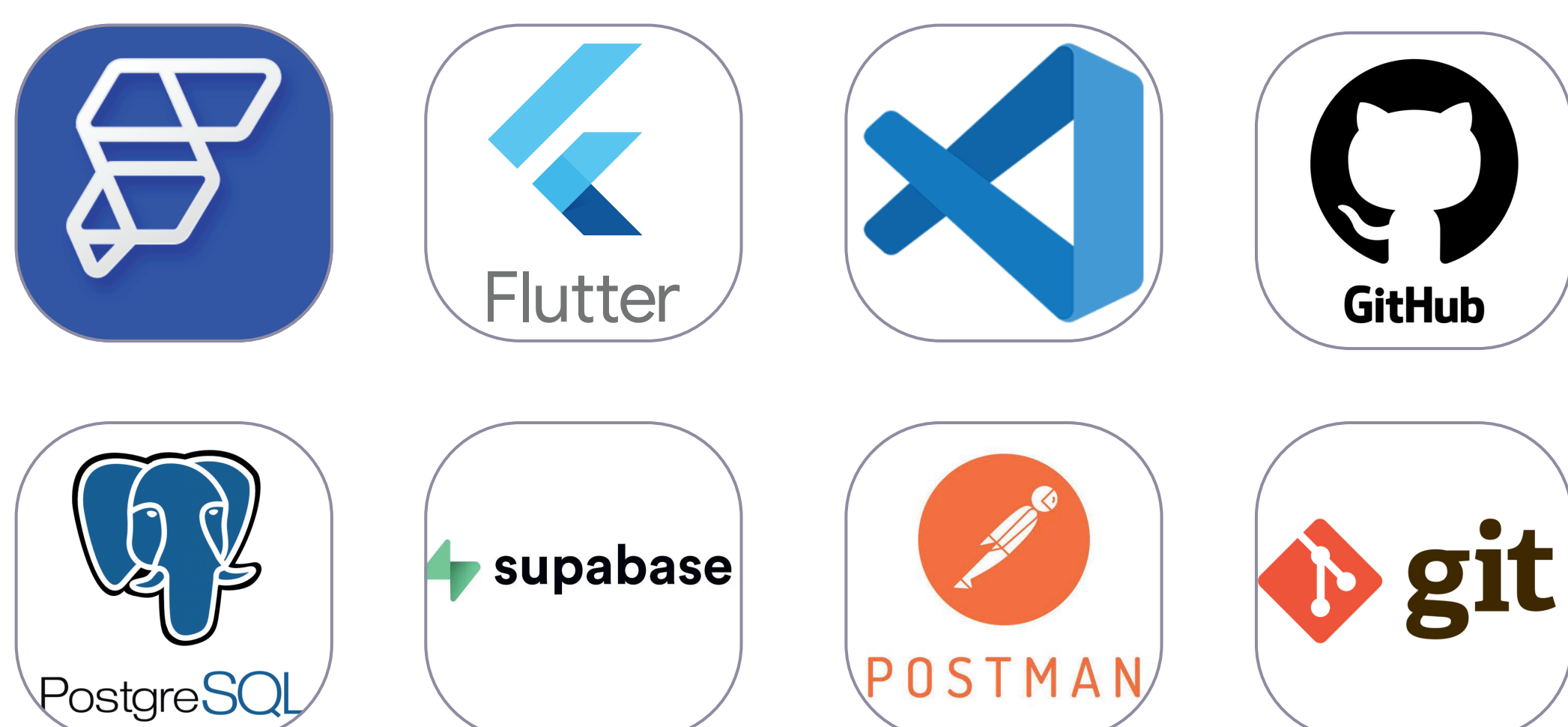


School Interface

Parent Interface

Bus Staff Interface

TOOLS



CONCLUSION

GateFlow provides an integrated solution that enhances school pickup and drop-off processes by improving communication, safety, and operational efficiency. The system replaces traditional manual methods with a unified digital platform that enables real-time updates and organized vehicle flow. By reducing congestion and minimizing delays, GateFlow creates a safer and more structured school environment for students, parents, and staff. Furthermore, the system aligns with smart transportation concepts by leveraging modern technologies to support digital transformation and improve overall school management.